From Algebra to Systems of Equations

Big question: How can equations help us find values we do not know?

1 Why do we use algebra?

Arithmetic helps us calculate with known numbers: **3 + 5 = 8**. Algebra helps us describe relationships and reason about numbers that are unknown or can change. We use letters called **variables** to represent those numbers.

An **expression**, such as x + 5, describes a quantity. An **equation**, such as x + 5 = 8, says that two quantities are equal. Solving an equation means finding a value that makes the equality true.

2 Start with one unknown

$$x + 5 = 8$$

What number plus 5 equals 8? Subtract 5 from both sides to preserve equality:

$$x + 5 - 5 = 8 - 5 \text{ so } x = 3$$

Check: 3 + 5 = 8. The answer works. In this example, one equation determines one unknown.

3 What happens when there are two unknowns?

$$x + y = 8$$

Now two numbers add to 8. Can we find both numbers from this information alone? Not uniquely. There are many pairs that work:

x	y	Check
1	7	1 + 7 = 8
3	5	3 + 5 = 8
5	3	5 + 3 = 8

Each ordered pair (x, y) is a solution to the equation. **One equation with two variables often leaves us with many possible answers.**

4 A second equation gives us another condition

Suppose we also know that x is 2 greater than y. We can write:

$$x + y = 8 \text{ and } x - y = 2$$

We need a pair that makes **both** equations true. Try (5, 3): 5 + 3 = 8 and 5 - 3 = 2. Both statements are true!

These two equations together are called a **system of equations**. A solution to the system must satisfy every equation at the same time.

5 Why are systems useful?

Real situations often give us more than one condition: the number of tickets sold and the money collected; the number of coins and their total value; or two travel plans that reach the same location. Systems let us combine those facts to find unknown quantities.

Important idea: To find a unique answer, we generally need enough *independent* information. Two variables and two equations can give one solution, but sometimes the equations have no common solution or infinitely many.

Discuss before learning a method

1. What is the difference between an expression and an equation?
2. Why does $x + y = 8$ have more than one solution?
3. Why must (x, y) satisfy both equations in a system?
4. Invent a real-world situation with two unknown quantities and two clues.

Next lesson: Show each equation as a line on a graph. Their intersection, if there is one, represents a pair that satisfies both equations.